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RESEARCH ARTICLE

True metabolizable energy of moist-soil seeds among dabbling duck species

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Abstract

True metabolizable energy (TME) assays were first developed to refine poultry diets but have since been adapted by wildlife biologists and conservation planners to determine the assimilable energy within natural wetland foods (i.e., seeds, invertebrates, vegetation) for waterfowl. Bioenergetic models incorporate TME values to quantify the energetic carrying capacity of wetlands. Prior TME research focused on the food items commonly found in mallard (*Anas platyrhynchos*) diets, leaving uncertainty in bioenergetics modeling as to the carrying capacity of other dabbling duck species in combination with a wider range of moist-soil plant seeds consumed by waterfowl. We collected 287 TME_N (nitrogen corrected TME) feeding trial samples between August and March 2020–2022 to quantify TME_N for 8 moist-soil plant seed species (*Cyperus erythrorhizos*, *Cyperus iria*, *Leptochloa panicoides*, *Panicum rigidulum*, *Polygonum hydropiperoides*, *Polygonum lapathifolium*, *Sesbania herbacea*, *Sida spinosa*) among 3 species of dabbling duck (mallard, northern pintail [*Anas acuta*], and green-winged teal [*A. crecca*]). We found TME_N values varied among duck and seed species (min–max = −0.29–2.58 kcal/g), with northern pintail values generally being 1–2 kcal/g lower than mallard and green-winged teal. Our results indicate that bioenergetics

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models could be improved by incorporating species-specific energy supply estimates, customized by species composition for the region or area comprising the planning unit. Moreover, management practices can be focused on providing plant species most beneficial to target species or groups of ducks representing the expected species composition in a management unit using this new information that illustrates the variability in assimilable energy across plant species and for dabbling duck species other than the generalist mallard.

KEYWORDS

Anas, dabbling duck, green-winged teal, mallard, moist-soil seeds, northern pintail, TME, true metabolizable energy, waterfowl

With losses and degradation of wetlands exceeding 90% in some regions of the United States, protection, restoration, and management of remaining wetlands are needed to support millions of migratory waterfowl and other wetland birds (Bellrose et al. 1983, Dahl 1990, Stafford et al. 2011). Considerable effort is expended annually on public and private lands to manage seasonally flooded wetlands for desirable natural seeds, tubers, and aquatic invertebrate production, while controlling undesirable plants to meet nutritional needs for non-breeding waterfowl (Soulliere et al. 2017, Brasher et al. 2018). Moist-soil wetlands are typically dewatered by natural or artificial drawdown during the spring or summer, providing exposed mudflats for the germination of annual, seed-producing plants that grow during late summer and early fall. Natural plants are then flooded during fall and winter to provide waterfowl access to seeds and provide substrate for invertebrate production (Fredrickson and Taylor 1982, Hagy and Kaminski 2012a). In managed seasonal-emergent wetlands, desirable plant species are those that provide energetic benefits through the production of large quantities of high energy foods consumed by waterfowl (Callicutt et al. 2011, Hagy and Kaminski 2012b). In contrast, undesirable plants are those that interfere with desirable plant growth and typically include perennial herbs, trees, shrubs, or species with invasive growth patterns that may produce cover but contribute little to waterfowl energetics due to limited seed production, avoidance by waterfowl, or low digestive efficiency (Hagy and Kaminski 2012a, Gross et al. 2020, Howard et al. 2023, Larson et al. 2024).

Emergent wetland managers are faced with a series of decisions that drive plant composition and often influence food availability for waterfowl. Wetland management decisions include the timing and duration of annual dewatering activities (i.e., drawdown), frequency of soil disturbance within and among years, and herbicide applications and physical vegetation manipulation during the growing season to limit invasive or otherwise undesirable species. Often, managers are limited by temporal, logistical, and financial constraints, but also a general lack of information contributes to uncertainty in evaluating tradeoffs to wisely expend resources to achieve their objectives under the North American Waterfowl Management Plan (NAWMP 2018, Hagy et al. 2022).

While some wetland management practices are driven by informal monitoring, recent advancements in science and the availability of rapid monitoring protocols provide reasonable approaches to more formalized adaptive resource management at scales needed by typical wetland managers (Hagy et al. 2021, 2022). Under an adaptive management framework, information on the energetic value (e.g., seed production potential, energy assimilability) of plant species and stand composition collectively can be used to drive management decisions within and across years in a more formal manner (Fleming et al. 2019, Gross et al. 2020, Martin et al. 2022, Howard et al. 2023, Larson et al. 2024). Furthermore, energetic value metrics are used by conservation planners to convert step-down continental population objectives associated with NAWMP to habitat objectives and estimate wetland protection and restoration goals (Fleming et al. 2019, Lancaster et al. 2021, Hagy et al. 2022, Lower Mississippi Valley Joint Venture 2022).

While previous energetic modeling work has relied on energetic assimilability estimates from domestic mallards (*Anas platyrhynchos*; Kaminski and Essig 1992, Sherfy 1999, Checkett et al. 2002, Dugger et al. 2007), recent science has suggested substantially more variation in energetic assimilability between duck and plant species than previously understood (Gross et al. 2020, Larson et al. 2024, Blenk 2025). For instance, Gross et al. (2020) documented differences in true metabolizable energy (TME) between mallards and gadwall (*Mareca strepera*) fed submersed aquatic vegetation. Additionally, when TME estimates for a particular plant species are unavailable for use in bioenergetic models, plant species-specific values are typically substituted with a mean value, which can result in substantial uncertainty around model estimates (Williams et al. 2014, Livolsi et al. 2015). Combined with a limited amount of information to describe energy availability and assimilability among multiple duck and plant species, wetland managers and conservation planners face numerous assumptions when evaluating management decisions, such as the assumption that managing for mallards benefits all waterfowl equally. To address the assumption that managing for the energetic needs of mallards benefits other waterfowl species, the U.S. Fish and Wildlife Service and collaborators have identified energetic assimilability and other bioenergetic model parameters as science priorities to advance management and planning for waterfowl (Soulliere et al. 2017, Lower Mississippi Valley Joint Venture 2022, U.S. Fish and Wildlife Service Southeast Region 2025).

Our objective was to estimate variation in true metabolizable energy (TME_N), which is a measure of assimilable energy of food items accounting for innate endogenous losses, among 8 moist-soil seed species and 3 dabbling duck species common in migration and wintering regions of the central United States. We hypothesized that TME_N would vary among duck and plant species, potentially complicating bioenergetic models used for conservation planning. Such information on variance among seed species would provide wetland managers with additional criteria to evaluate plant composition, but variation among duck species would complicate wetland management decisions, necessitating additional species-specific objectives or other criteria in conservation planning models.

STUDY AREA

We conducted TME_N trials at the Forbes Biological Station located on Chautauqua National Wildlife Refuge near Havana, Illinois, USA (40.353° N, 90.022° W; 145 m elevation) during August–March 2020–2022. The average high temperature was 31.7°C in August, and the average low temperature was −8.9°C in January (U.S. Climate Data 2020).

METHODS

Waterfowl capture and husbandry

We used wild-caught green-winged teal (*Anas crecca*), northern pintail (*A. acuta*), and mallards to ensure TME estimates were representative of wild individuals (Kaminski and Essig 1992, Gross et al. 2020, Larson et al. 2024). We captured green-winged teal, northern pintail, and mallards in Illinois and western Tennessee, USA, using y-confusion traps, swim-in traps, and rocket nets (Sharp and Smith 1986, Dieter et al. 2009). Ducks were acclimated to captivity for ≥30 days before use in trials and released following the conclusion of the study. Outside of TME_N trials, all birds were confined in unheated outdoor pens (2.0 m × 6.0 m × 2.0 m), subject to natural temperature and photo period while provided with *ad libitum* fresh water and commercial game bird ration (Purina 30% Protein Starter Game Bird Feed, crude protein 30%, crude fat 2.50%, crude fiber 6.5%). We did not provide grit while in captivity or during the trial period (Bilgili et al. 1982, Sherfy et al. 2005).

Seed collection and processing

We collected seeds from 8 moist-soil species (*Cyperus erythrorhizos*, *C. iria*, *Polygonum hydropiperoides*, *P. lapathifolia*, *Panicum rigidulum*, *Sesbania herbacea*, *Leptochloa panicoides*, and *Sida spinosa*) from Chickasaw National Wildlife Refuge (NWR) near Halls, Tennessee (35.514° N, 89.334° W), Chautauqua NWR near Havana, Illinois, USA (40.353° N, 90.023° W), Hatchie NWR near Brownsville, Tennessee (35.500° N, 89.26° W), and Overflow NWR near Wilmot, Arkansas, USA (33.095° N, 91.64° W). The locations were chosen due to collaboration with several researchers on this study through companion projects (Bethell 2022, Martin et al. 2022) and represent the broad geographical range of the seed species within this study. We chose seed species previously reported as consumed by dabbling ducks in the literature and identified through a companion study (Bethell 2022), to reduce duplication with previous TME studies (McClain 2017), and of known interest to wetland managers across a large portion of the central and eastern United States (Swanson and Bartonek 1970, Swanson et al. 1974, Hagy and Kaminski 2012a, Tidwell et al. 2013, Klimas et al. 2022). We separated seeds from other vegetation material (e.g., leaves, stems) using a series of sieves of descending sizes (#15 [1.25 mm], #35 [0.50 mm], #60 [0.25 mm], and #100 [0.15 mm]). All seeds were stored indoors at room temperature.

Fasting trials

To minimize fasting time and ensure all foods had been voided from an individual duck's digestive system, we performed a titration trial prior to the main trial period. We randomly selected 12–16 individuals of each duck species, removed access to commercial ration, and collected excreta every 6 hr during a 48-hr fasting period ($n = 8$ excreta samples for each duck; Parsons et al. 1982, Wang et al. 2010, Larson et al. 2024). Birds were supplied with access to water *ad libitum*, and excreta collection and processing methods followed the feeding trials described herein.

Feeding trials

We conducted feeding trials between August and March of 2020–2022 following general procedures outlined in Gross et al. (2020). At the beginning of a TME_N trial, 5–20 individuals (mixed sexes) were weighed and randomly assigned to either the treatment or control group. Individuals were excluded from trials if body mass was <85% of their mass at capture or if they failed to gain >85% of mass lost during a previous trial. Then, individuals were placed in individual fasting cages and provided water *ad libitum* for 12–24 hr, depending on species (Figure 1). After fasting, we attached harnesses with a whirl-pak® (Whirl-Pak, Nasco Sampling, Pleasant Prairie, WI, USA) affixed around the duck's cloaca to collect excreta (Lancaster et al. 2019). For control birds, we only collected fecal material and did not feed individuals trial foods. Initially, we designed the study with the intent of using individual ducks in multiple TME_N trials to serve as their own control (Sherfy et al. 2005). However, some individuals, especially green-winged teal, failed to maintain or regain body mass while being held in captivity and undergoing TME_N trials. As a result, we switched to group controls, which allowed us to catch large numbers of green-winged teal and subject them to trials only once and then release them (Larson et al. 2024). While mallard and northern pintail individuals did undergo multiple trials, as originally designed, we still used group controls for those species for consistency (Sherfy et al. 2005, McClain 2017). For the group control, we averaged the energy from the control excrement samples among all birds of a given species (individuals within the control group, mallard = 32, green-winged teal = 17, northern pintail = 27).

We fed each treatment bird the equivalent of ~1% of its body mass of a single seed species by inserting an open-ended 3-ml syringe preloaded with a known dry mass (0.25 g, 0.50 g, 1.0 g, 2.0 g, 3.0 g) of a single seed species into the esophagus and pushing the food out of the syringe with the rubber plunger (0.9 cm x 5.0 cm; Adeola et al. 1997, Lancaster et al. 2019, Gross et al. 2020).

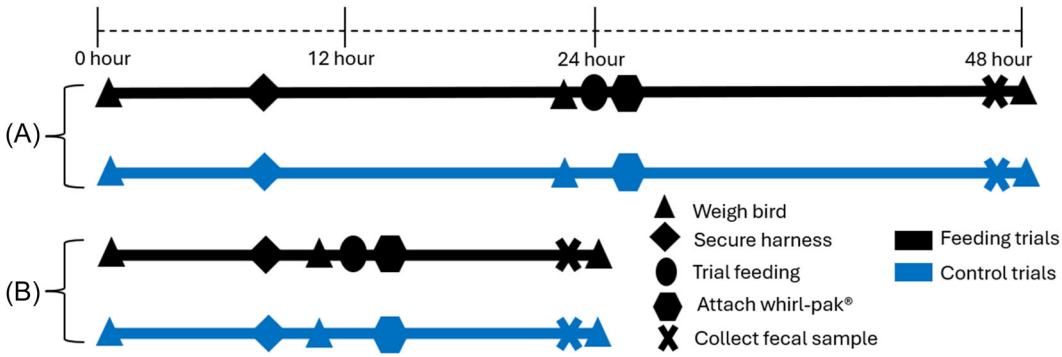


FIGURE 1 Timeline of true metabolizable energy (TME_N) trials at the Forbes Biological Station in Havana, Illinois, USA, from 2020-2022. The black lines represent the feeding trials, and the blue lines represent the control trials. (A) Larger duck species (mallards [*Anas platyrhynchos*] and northern pintail [*A. acuta*]) were on a 24-hour fasting period followed by a 24-hour fecal collection period. (B) The smaller green-winged teal (*A. crecca*) was on a 12-hour fasting period and then a 12-hour fecal collection period.

Combinations of prepared syringes were used to feed the correct dry mass for each seed species and bird combination. We replicated inserting the syringe with seeds into the esophagus of each bird 4–10 times depending on the individual bird mass (i.e., green-winged teal required 2–4 syringes; northern pintail and mallards required 6–10 syringes). We ensured all feeding times were <5 min/individual. Following feeding, each individual bird (both treatment and control) was placed in an individual metabolic cage to restrict movement, reduce wasted energy, and collect any excreta that might have leaked from the whirl-pak® harness (Lancaster et al. 2019). During this fecal collection period, we provided all ducks with *ad libitum* water. Individual birds used in multiple trials were rested for ≥14 days to recover any lost body mass before use in a subsequent trial.

Laboratory analysis

After excreta collection, samples were frozen for 30–450 days before sample preparation for analysis (removing any feathers or other contaminants from the excreta sample). We dried the remaining excreta in an oven at 80°C for ≥48 hr, weighed the sample to the nearest 0.0001 g, and ground it with a mortar and pestle. We measured the gross energy of each moist-soil seed species (GE_F, kcal/g), energy of excreta from treatment ducks (EE_F), and energy of excreta from control ducks (EE_C) in duplicate samples using an adiabatic oxygen bomb calorimeter (30 atm O_2; Parr Instrument Company, Rock Island, IL, USA). We formed and compressed 2 subsamples of excreta (0.5 g) into pellets and subjected them to the calorimeter. A mineral oil spike was used to ensure thorough combustion of the sample. We used Parr Instrument Company calorimeter software to calculate the energy released from a sample after the combustion of the sample (Gross et al. 2020). Seed and excreta samples were further analyzed at the University of Illinois Urbana-Champaign Animal Science Lab to determine percent water and nitrogen-free extract using standard methods (Servello et al. 2005). Percent water was determined by drying samples in a forced air oven at 110°C for 24 hr, and percent nitrogen was calculated using a Carlo Erba NA Thi1500 Elemental Analyzer (Champaign, Illinois, USA).

TME_N calculation

To account for potentially greater catabolism of body tissue by control ducks and avoid overestimating energy derived from non-food origins, we corrected TME to zero nitrogen balance (TME_N;

Parsons et al. 1982, Sibbald and Morse 1983, Gross et al. 2020). We calculated TME_N (kcal/g) using the following equation:

$$TME_N = \frac{((DM_F)GE_F) - [(EE_F + 8.73(DM_F * \%N_F - DM_E * \%N_E)) - (EE_C + 8.73(DM_{CF} * \%N_F - DM_{CE} * \%N_{CE}))]}{DM_F}$$

Details on each individual variable in the TME_N equation can be found in Table 1.

Statistical analysis

We determined the appropriate fasting time for each species by testing for differences in excreta mass collected among the 6-hr periods using a one-way repeated analysis of variance and Tukey's pairwise comparisons ($\alpha = 0.05$). To determine whether TME_N differed among bird species and seed species, we fit a set of linear mixed-effects models (Table 2). These models included a model with an interaction term between the seed species and bird species, as well as single-variable models and models with all 3 fixed effects: 1) seed species, 2) bird species, and 3) cumulative trial number for that individual, which ranged from 1–7 trials. We included the cumulative trial number to examine whether repeated trials on the same individual influenced TME_N values. We included individual duck ID as random effects to account for repeated measures of individuals (Zuur et al. 2010).

TABLE 1 Variables used in the TME_N calculation for duck feeding studies at the Forbes Biological Station in Havana, Illinois, USA during 2020–2022, with a definition, units, and how they were calculated.

Variable Name	Definition	Units	How Calculated
DM_F	Total dry mass fed	g	mass fed multiplied by the % dry mass fed
GE_F	Gross energy of the food item	kcal/g	
EE_F	Energy Excreted from a treatment bird	kcal/g	Gross energy of the fecal sample divided by the dry mass of the fecal sample
8.73	Energetic constant of urinary nitrogen compounds in fecal sample	kcal/g	Titus et al. (1959)
$\%N_F$	Percent Nitrogen in Trial Food	%	Determined using a Carlo Erba NA Thi 1500 Elemental Analyzer at the Animal Science Lab
DM_E	Dry Mass of excreta	g	Mass of excreta multiplied by the % dry mass of excreta
$\%N_E$	Percent Nitrogen in Excreta	%	Determined using a Carlo Erba NA Thi 1500 Elemental Analyzer at the Animal Science Lab
EE_C	Energy excreted from a control bird	kcal/g	We used the average energy excreted by control birds for each duck species to estimate EE_C
DM_{CF}	Dry mass fed to control	g	This is always 0
DM_{CE}	Dry mass of control bird excreta	g	Mass of excreta multiplied by the % dry mass of control bird excreta
$\%N_{CE}$	Percent Nitrogen in the control bird excreta	%	Determined using a Carlo Erba NA Thi 1500 Elemental Analyzer at the Animal Science Lab

TABLE 2 Model, number of parameters (*K*), Akaike's Information Criterion (AIC_c), the difference in AIC_c score from the highest-ranked models (ΔAIC_c), and model weights (ω_i) from all univariate and competitive multivariate linear mixed models that examined variation in true metabolizable energy (TME_N) of 6 moist-soil seed species (*Cyperus erythrorhizos*, *Cyperus iria*, *Panicum rigidulum*, *Polygonum lapathifolium*, *Polygonum hydropiperoides*, and *Sesbania herbacea)* among 3 duck species (mallard, *Anas platyrhynchos*; northern pintail, *A. acuta;* and green-winged teal, *A. crecca)* at Forbes Biological Station located in Havana, Illinois, USA, during 2020–2022. Random effects within the model were (1|Duck ID).

Model Fixed Effects	*K*	AIC_c	ΔAIC_c	ω_i
Seed species ∗ duck species	20	1004.26	0.00	0.90
Seed species + duck species	10	1012.34	8.08	0.01
Seed species + duck species + individual trial[a]	11	1012.98	8.72	0.01
Duck species	5	1021.29	17.03	0.00
Seed species	8	1042.047	37.78	0.00
Null	3	1050.80	46.54	0.00

[a]Individual trial represents the cumulative trial number the individual duck was on (min–max = 1–7 trials per bird).

We performed model selection using Akaike's Information Criterion corrected for small sample sizes (AIC_C; Burnham and Anderson 2002; Table 2). We considered models within $\Delta AIC_C \leq 2$ of the top model as competitive (Arnold 2010). We quantified 85% confidence intervals surrounding model estimates, and considered parameter estimates different if confidence intervals did not overlap zero. We report marginal and conditional R^2 as measures of goodness-of-fit (Nakagawa and Schielzeth 2013). All statistical analyses were completed in R version 4.3.2 (R Core Team 2024) and used the lme4 package to fit our models (v1.4-34; Bates et al. 2015).

RESULTS

We found significant variation in excreted mass among 6-hr periods for all 3 duck species (Figure 2; mallard $F_{8,\ 120} = 45.85$, $P < 0.001$; northern pintail $F_{8,\ 120} = 102.9$, $P < 0.001$; green-winged teal $F_{8,\ 120} = 173.7$, $P < 0.001$). A post-hoc Tukey test showed a decrease in excreted mass between hours 6 and 24 for mallards and northern pintail, and a decrease in excreted mass between hours 6 and 12 for green-winged teal. We used these results in our trial study design, fasting mallards and northern pintails for 24 hours and green-winged teal for 12 hours.

In total, we collected 287 samples among 8 moist-soil plant seed species and 3 dabbling duck species (Table 3). On average, we collected 16 samples per seed and duck species combination (min–max = 8–23; Table 3). We excluded data from birds that regurgitated any seeds following the feeding process (~10% of fed ducks). We also excluded 2 seed species from our model (*Leptochloa panicoides* and *Sida spinosa*), as we were unsuccessful in obtaining samples across all 3 duck species. We excluded those seeds from our modeling effort to have a balanced design (all seeds fed to all duck species), but we do report all seed species raw TME_N values and gross energy for all 8 seed species (Table 3).

We had a singular competitive model (seed species ∗ duck species) that accounted for 90% of the total model weight (Table 2). The fixed effects of seed and duck species explained 27.6% of the variation in TME_N (Conditional R^2), whereas the combination of fixed and random effects of individual ID explained 31.2% of the variation in TME_N (Marginal R^2).

We found an interaction where estimates of most seed and duck species combinations differed from zero. For most seed species, mallard and green-winged teal had similar TME_N values that were both greater than those of northern pintail (Table 3). The exceptions were *Cyperus iria*, and *Sesbania herbacea* (Table 3). Notably, *Sesbania*

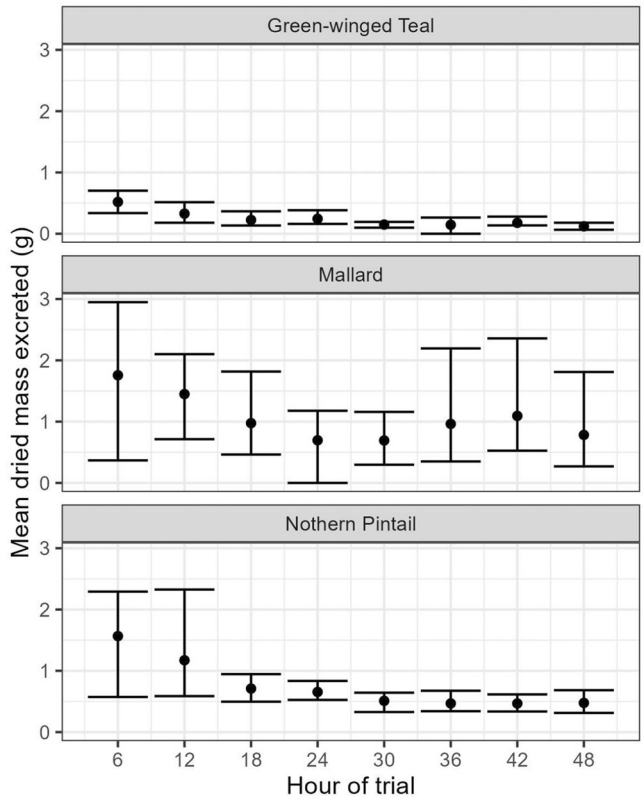


FIGURE 2 Mean dry mass and 95% confidence intervals of excreta collected at 6-hr intervals during fasting titration trials for mallard (*Anas platyrhynchos*), green-winged teal (*A. crecca*), and northern pintail (*A. acuta*) at the Forbes Biological Station in Havana, Illinois, USA, during 2020–2022.

herbacea TME_N was lowest of any plant species for green-winged teal and mallard, although confidence intervals overlapped between *Sesbania herbacea* and several other species for mallard (Figure 3; Tables 3 and 4).

DISCUSSION

As predicted, seed and duck species were predictors of TME_N, providing wetland managers with additional information to help them tailor vegetation communities to their targeted duck communities. Generally, all moist-soil seed species had TME_N in the range reported by other studies for moist-soil seed species (Hagy and Kaminski 2012a, Gross et al. 2020, Blenk 2025), but all estimates were below the fixed value of 2.5 kcal/g commonly used by conservation planners (Reinecke et al. 1989, Checkett et al. 2002, Kaminski et al. 2003). Similarly, McClain (2017) noted that the mean TME_N estimate of seeds from moist-soil plants was 1.78 kcal/g, substantially below the previously suggested average. Marty (2017) estimated a 1–5% reduction in carrying capacity of idled and production ricelands of the Gulf Coast using species-specific biomass and TME estimates, versus the average 2.5 kcal/g for moist-soil seeds. Given the expanding literature basis reporting TME values across a range of plant and seed species, conservation planners may want to reexamine bioenergetic models in their planning efforts that use fixed TME or energy supply estimates derived using these estimates (Williams et al. 2014). This study provides additional evidence that managing plant composition in seasonal emergent wetlands is an important consideration

TABLE 3 Model-predicted true metabolizable energy (TME_N; kcal/g) estimates with lower and upper 85% confidence intervals for the 6 seed species fed to mallard (*Anas platyrhynchos*), green-winged teal (*A. crecca*), and northern pintail (*A. acuta*) at Forbes Biological Station located in Havana, Illinois, USA, during 2020–2022.

				85% Confidence Interval	
Seed species	**Duck species**	***n***	**TME_N**	**Lower**	**Upper**
Redroot flatsedge (*Cyperus erythrorhizos*)	Mallard	22	2.16	1.75	2.65
	Green-winged teal	11	2.58	2.04	3.14
	Northern pintail	19	0.33	−0.12	0.76
Marsh pepper (*Polygonum hydropiperoides*)	Mallard	16	2.09	1.66	2.64
	Green-winged teal	14	2.01	1.61	2.41
	Northern pintail	16	0.33	−0.15	0.78
Nodding smartweed (*Polygonum lapathifolia*)	Mallard	10	1.91	1.28	2.54
	Green-winged teal	8	1.43	0.80	2.06
	Northern pintail	18	0.40	−0.08	0.73
Ricefield flatsedge (*Cyperus iria*)	Mallard	13	1.92	1.45	2.46
	Green-winged teal	15	1.33	0.87	1.73
	Northern pintail	18	0.97	0.56	1.40
Redtop panicgrass (*Panicum rigidulum*)	Mallard	14	1.30	0.82	1.75
	Green-winged teal	14	1.50	1.04	1.99
	Northern pintail	23	0.03	−0.41	0.44
Coffeeweed (*Sesbania herbacea*)	Mallard	22	1.08	0.71	1.54
	Green-winged teal	14	−0.29	−0.69	0.28
	Northern pintail	20	0.68	0.31	1.20

for wetland managers, especially when considered in tandem with overall moist-soil plant species seed productivity (Howard et al. 2023), to help achieve their energetic carrying capacity goals for a suite of duck species rather than assuming static values or planning using only a single species (e.g., mallard).

Interestingly, northern pintail derived less energy from most seed species than did mallards or green-winged teal, with the exception of *Sesbania herbacea*. While mallard and green-winged teal TME_N values generally were within previously reported ranges for dabbling ducks, our northern pintail values were some of the lowest seed TME_N estimates for this species (Hoffman and Bookout 1985, Ballard et al. 2004). Of note, we did not include seed species that had been previously estimated for northern pintail, but *Polygonum* sp. congeners were similar between our study and Hoffman and Bookout (1985). Very little previous work has been done with northern pintail TME_N assays compared to mallard, but Miller (1984) showed that this species assimilated poultry ration and 1 moist-soil seed species similar to other dabbling duck species. It is possible that northern pintail compensate for low energy assimilability through selection methods or by choosing alternative species when available. In a related project, Bethell (2022) noted significant differences in frequency of occurrence and selection tendency of seed species in the diets of mallards, northern pintail, and green-winged teal. For example, Bethell (2022) reported very high proportional occurrence of *Panicum rigidulum* only in the diet of northern pintail. In contrast, they reported high occurrence of *Cyperus erythrorhizos* in green-winged teal, low occurrence in mallards, and no occurrence by

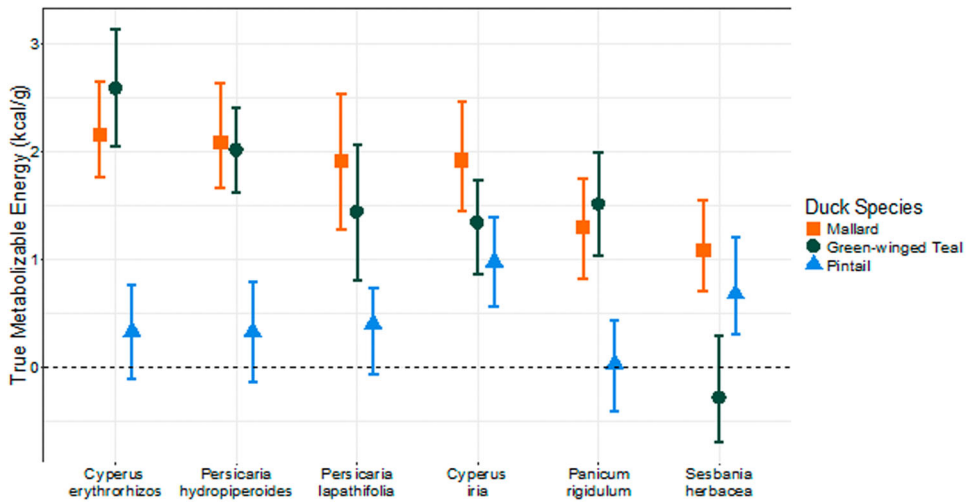


FIGURE 3 Model-predicted true metabolizable energy (TMEN; kcal/g) estimates with lower and upper 85% confidence intervals for the 6 seed species fed to mallard (*Anas platyrhynchos*), green-winged teal (*A. crecca*), and northern pintail (*A. acuta*) at Forbes Biological Station located in Havana, Illinois, USA, during 2020–2022.

northern pintail. The combined factors of seed size, energy content, assimilability, and perhaps other characteristics likely affect diet (i.e., seed selection) differentiation as species forage in the same places. A similar trade-off may extend to digestion efficiency, possibly through mechanisms such as genetics and gut microbiome (e.g., Nudds et al. 2000, Wen et al. 2021).

Sesbania herbacea is a common and prolific seed-producing plant species in moist-soil wetlands in the central and southeastern United States (Hagy and Kaminski 2012a; b). However, there is limited evidence that *Sesbania herbacea* is regularly consumed by dabbling ducks despite its abundance in moist-soil wetlands (Hagy and Kaminski 2012a, Bethell 2022). Our results indicating a low TME_N across all 3 duck species may help explain avoidance of *Sesbania herbacea* by dabbling ducks while foraging (Bethell 2022). Whether to control *Sesbania herbacea* in moist-soil wetlands is a common question among wetland managers, and mounting evidence suggests *Sesbania herbacea* has little to no foraging value for waterfowl (e.g., Bethell 2022). Given that the vegetation growth of this species can also quickly outcompete other more desirable moist-soil species, there appears to be no obvious energetic benefits for dabbling ducks from *Sesbania herbacea* in a moist-soil wetland, suggesting managers should limit coverage in lieu of more beneficial species.

Working with birds in captivity was necessary for our study to be able to control a variety of variables while estimating TME_N values, but we acknowledge introduction of other potential bias from this design. Our study used wild-caught adult ducks with the assumption that they would best represent the wild population (Gross et al. 2020, Beach et al. 2024, Larson et al. 2024). However, captivity leads to stress in many wild animals, both short-term and long-term, which likely has unknown effects on digestive efficiency (Morgan and Tromborg 2007, Fischer and Romero 2018, Beach et al. 2024). While we acknowledge potential effects of stress and other physiological factors on TME_N estimates, our results are comparable to previous research and used advancements advocated by recent research for process and husbandry improvements (Lancaster et al. 2019, Gross et al. 2020). Given the intensive nature of TME_N methods (e.g., captive flock care, potential effects of stress in captivity) and availability of published TME_N estimates for duck and seed species (Petrie et al. 1998, Checkett et al. 2002, Coluccy et al. 2015, Larson et al. 2024, Blenk 2025), future work could explore estimating TME_N values without captive bird trials (Blenk 2025).

An additional advancement in TME_N estimation techniques is better understanding of excreta voidance dynamics across duck species. Using the minimum fasting time possible ensures that most undigested food is

TABLE 4 Raw true metabolizable energy (TME_N; kcal/g) estimates with standard error, sample sizes (*n*), and gross energy (GE; kcal/g) for the 6 seed species fed to mallard (*Anas platyrhynchos*), green-winged teal (*A. crecca*), and northern pintail (*A. acuta*), including 2 additional seed species not fed to all 3 duck species (*Leptochloa panicoides* and *Sida spinosa*). All data were collected at the Forbes Biological Station located in Havana, Illinois, USA, during 2020–2022.

Seed species	GE	Duck species	*n*	TME_N
Redroot flatsedge (*Cyperus erythrorhizos*)		Mallard	22	2.16 ± 0.17
	4.29	Green-winged teal	11	2.56 ± 0.22
		Northern pintail	19	0.70 ± 0.37
Marsh pepper (*Polygonum hydropiperoides*)		Mallard	16	2.09 ± 0.16
	4.05	Green-winged teal	14	2.01 ± 0.14
		Northern pintail	16	1.28 ± 0.64
Nodding smartweed (*Polygonum lapathifolia*)		Mallard	10	1.93 ± 0.22
	4.13	Green-winged teal	8	1.44 ± 0.48
		Northern pintail	18	1.02 ± 0.45
Ricefield flatsedge (*Cyperus iria*)		Mallard	13	1.92 ± 0.16
	3.98	Green-winged teal	15	1.33 ± 0.44
		Northern pintail	18	0.95 ± 0.29
Redtop panicgrass (*Panicum rigidulum*)		Mallard	14	1.30 ± 0.16
	3.56	Green-winged teal	14	1.50 ± 0.22
		Northern pintail	23	0.28 ± 0.29
Coffeeweed (*Sesbania herbacea*)		Mallard	22	1.08 ± 0.20
	4.25	Green-winged teal	14	−0.3 ± 0.98
		Northern pintail	20	0.91 ± 0.33
Sprangletop (*Leptochloa panicoides*)		Mallard	16	1.96 ± 0.12
	3.91	-	-	-
		Northern pintail	16	1.16 ± 0.57
Prickly sida (*Sida spinosa*)		Mallard	9	2.04 ± 0.22
	4.23	-	-	-
		-	-	-

passed, which will reduce stress, body tissue catabolism, and help birds maintain better body condition more comparable to wild conditions under realistic short-term fasting conditions (e.g., migration days, freeze ups, etc.). In this study, there was clear evidence that green-winged teal and northern pintail voided most of their digested food in 12–18 hr while mallards voided their fed sample by 24 hr. Similarly, Larson et al. (2024) showed that lesser scaup (*Aythya affinis*) fasting periods should be no longer than 12–18 hr. Thus, future researchers should use shorter, faster periods (i.e., 12–24 hr instead of 48 hr; Checkett et al. 2002) that minimize tissue catabolism.

As with other recent publications using wild-caught ducks (Gross et al. 2020, Larson et al. 2024), we found substantial variation around TME_N estimates, which exceeded earlier work with game farm or domestic stock ducks

(Petrie et al. 1998, Checkett et al. 2002, Dugger et al. 2007, Coluccy et al. 2015). As noted by Gross et al. (2020), we suspect that using wild-caught birds likely increases the realism of TME_N estimates but may introduce additional variation through differences in individual metabolism, gut morphology, microbiota, and stress. We echo previous recommendations to increase experimental diet amounts, decrease fasting times, more precisely account for endogenous losses, and explore nutrient supplements during the trial period to reduce variation in TME_N estimates (Gross et al. 2020).

Our results show that mean TME_N is variable across both duck and plant seed species, although the high level of variation among individual ducks probably ameliorates some of the effective differences for bioenergetic models. While our recommendations for future studies should help improve husbandry practices and precision around TME estimates, recent research with wild-caught ducks has revealed a great deal more individual variation around ducks than was apparent in previous research conducted on domestic lineages. We suspect that the precise estimates from domestic lineage individuals are likely less realistic than observed in our study. Increased variation suggests a practical inability to deal with these nuances in rather simple bioenergetic models with current science. However, bioenergetics models are still useful to evaluate management of emergent wetlands and for estimating and evaluating the energetic needs for a variety of waterfowl in conservation planning. We suggest future work estimating TME_N do so across multiple duck species whenever possible so that wetland management can be informed, considering the needs of waterfowl more broadly.

MANAGEMENT IMPLICATIONS

Our results help address the direct science needs of the conservation planning and wetland management communities. Given apparent variation in assimilable energy among dabbling duck species, we recommend that conservation planners continue moving towards species-specific objectives and more complex parameterization of bioenergetics models that more realistically reflect the natural variation among duck and plant species. Continued use of TME_N values that do not represent the full breadth of duck and plant species (e.g., mallards to represent all dabbling ducks, grouped data values calculated from a few moist-soil seed species) could lead to oversimplified models that do not produce realistic results. Using rapid monitoring techniques for food production in waterfowl-managed wetlands (moist-soil or grain production) has greatly increased efficiency in the energy supply estimation process; wetland managers can quickly and efficiently use our information to make decisions under a transparent adaptive resource management framework.

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CONFLICTS OF INTEREST STATEMENT

The authors declare no conflicts of interest.

ETHICS STATEMENT

All work was done in accordance with University of Illinois Urbana Champaign's Institutional Animal Care and Use Committee protocol No. 21105, along with state and federal collection permits (TWRA Scientific Collection Permit No. 2358, IDNR Scientific Collection Permit No. W19.6079, and USFWS collection permit No. MB62822D-0).

DATA AVAILABILITY STATEMENT

The data that support the findings of this study are available from the corresponding author upon reasonable request.

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